

Design and Implementation of a Redundant Network Infrastructure

FortiGate High Availability with FortiLink-Managed MC-LAG Switching

Engineering Implementation Report

Physical laboratory implementation

2 × FortiGate 100E · 2 × FortiSwitch 1048E

FortiOS 7.2.13

July 2026

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1. Abstract

This report documents the design, implementation and verification of a redundant network infrastructure built on Fortinet hardware in a physical laboratory environment. The infrastructure comprises two FortiGate 100E firewalls configured as an Active-Passive High Availability cluster and two FortiSwitch 1048E switches operated as a Multi-Chassis Link Aggregation peer group under centralised control of the firewall cluster.

The engineering objective was the systematic elimination of single points of failure across two distinct layers of the network. High Availability addresses failure of the security gateway; MC-LAG addresses failure of the access switching layer; FortiLink binds the two into a single administrative and control domain so that a failover at the firewall tier does not orphan the switching tier. The report argues that these three technologies are not independent features deployed alongside one another, but a set of mutually dependent design decisions, none of which delivers end-to-end redundancy in isolation.

The completed implementation was verified through the FortiGate graphical interface and command line interface. The cluster reports both members synchronised with identical configuration checksums, both FortiSwitch units authorised and operational under FortiLink, and the Inter-Chassis Link established between the two switch chassis. Configuration and connectivity faults encountered

during deployment, together with the diagnostic reasoning used to resolve them, are documented in full.

2. Introduction

In a conventional single-firewall topology, the security gateway is simultaneously the most privileged and the most fragile element of the network. All north-south traffic traverses it, all policy is enforced at it, and its failure is total: no alternative forwarding path exists. The same argument applies one layer down. Where access switches are deployed as independent, singly-connected units, the loss of one switch severs every endpoint attached to it regardless of how much redundancy exists above.

Redundancy at only one of these layers produces an infrastructure whose availability is governed by whichever layer remains unprotected. A redundant firewall pair behind a single switch is no more available than the switch. A redundant switch pair beneath a single firewall is no more available than the firewall. Meaningful availability therefore requires that both layers be made redundant, and further requires that the redundancy mechanisms at each layer be aware of one another.

This project implements that principle on physical Fortinet hardware. The work was carried out entirely on production-grade equipment in a laboratory environment; no part of the infrastructure was simulated or emulated. All configuration, cabling, verification and fault resolution described in this report was performed directly on the devices.

3. Project Objectives

The implementation was carried out against five stated objectives:

- **Eliminate single points of failure.** Eliminate single points of failure across the firewall and switching layers of the infrastructure.
- **Firewall redundancy.** Provide firewall redundancy such that the failure of one security gateway does not interrupt the forwarding path or discard established sessions.
- **Centralised switch management.** Provide centralised switch management so that both access switches are administered as extensions of the firewall cluster rather than as separately managed devices.
- **Resilient Layer 2 architecture.** Build a resilient Layer 2 switching architecture in which redundant paths are simultaneously active rather than held in a blocked standby state.
- **Improve availability and reliability.** Improve overall network availability and reliability by ensuring that the redundancy mechanisms at each layer interoperate as a single system.

4. System Architecture

The architecture is a two-tier design. The upper tier is a FortiGate 100E cluster operating in Active-Passive mode, presenting a single logical security gateway to the network. The lower tier is a pair of FortiSwitch 1048E chassis operating as an MC-LAG peer group, presenting a single logical link aggregation partner to the cluster above.

The two FortiGate units are joined by dedicated heartbeat links and each is connected to both FortiSwitch chassis. The two FortiSwitch units are joined to one another by an Inter-Chassis Link. The result is a topology in which no single device and no single cable constitutes a cut vertex of the forwarding graph.

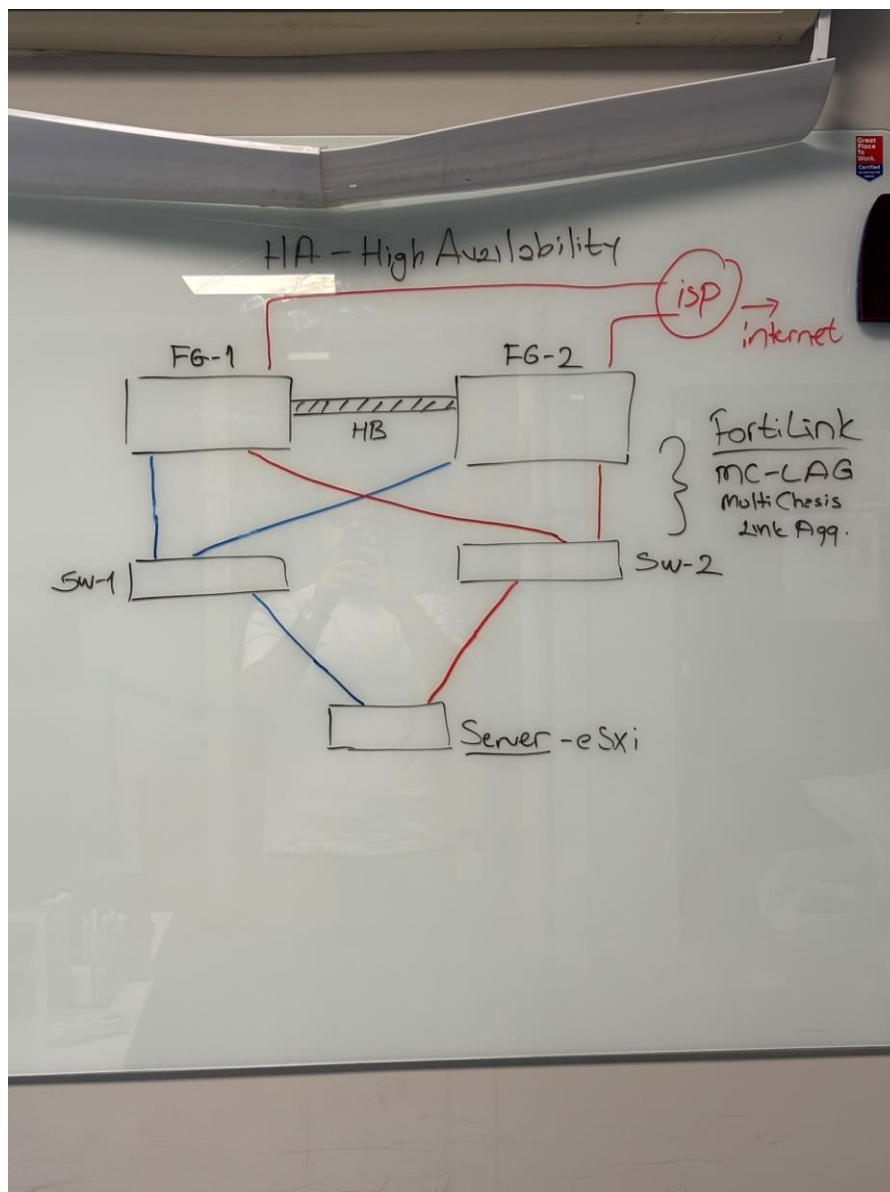


Figure 1. Original network design produced during the planning phase, showing the FortiGate pair with heartbeat interconnect, cross-connection to both FortiSwitch units, the ISP uplink and the dual-homed ESXi host.

Figure 2 restates the same design in schematic form, annotated with the interface assignments and addressing established during implementation.

FortiGate HA Cluster with FortiLink-Managed MC-LAG Switching

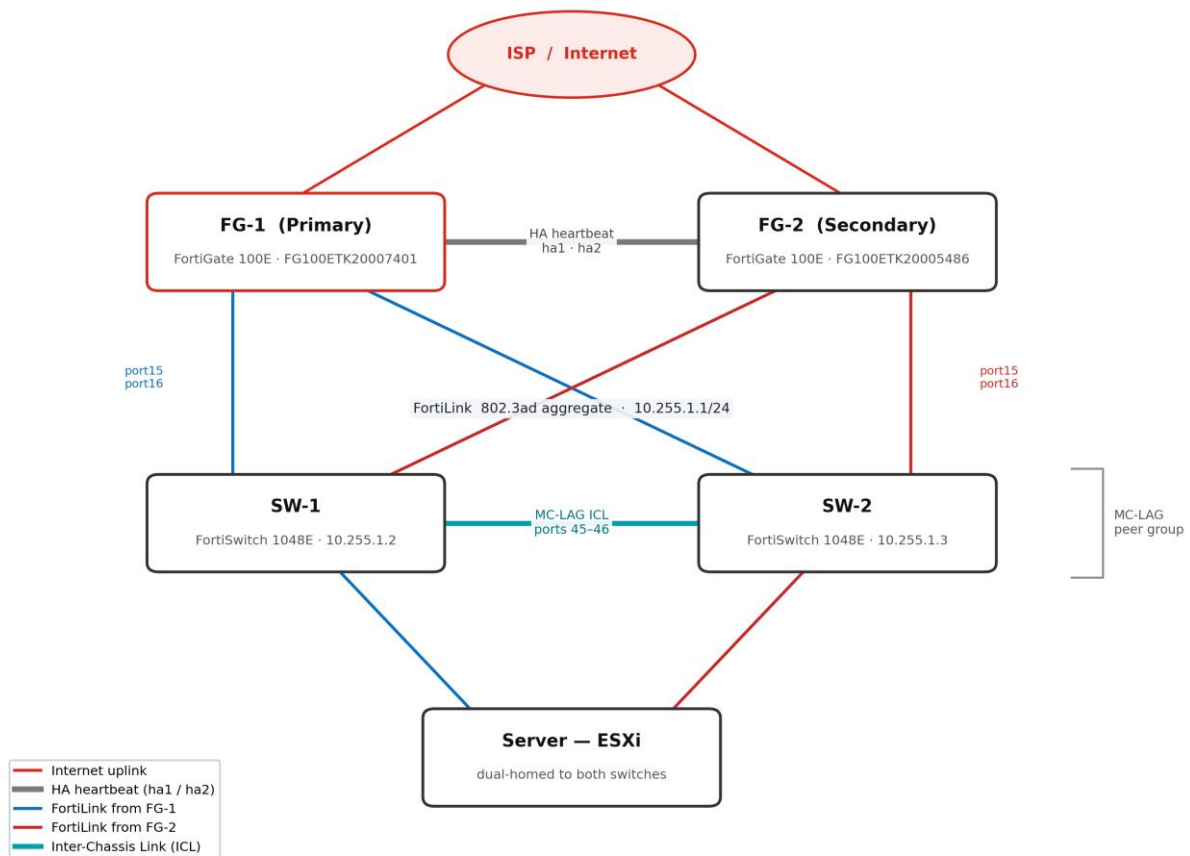


Figure 2. Implemented topology. The FortiLink aggregate carries the 10.255.1.1/24 management network; the switches occupy 10.255.1.2 and 10.255.1.3 within it. The ICL between the two chassis uses ports 45–46.

4.1 Traffic and control paths

Two logically distinct flows share this physical topology. The data path carries production traffic from endpoints, through the switch pair, into the active FortiGate, and onward to the ISP uplink. The control path carries FortiLink management traffic between the firewall cluster and the switches over the 10.255.1.0/24 network, together with HA heartbeat and configuration synchronisation traffic between the two cluster members over the dedicated ha1 and ha2 interfaces.

Separating the control path onto dedicated heartbeat interfaces is deliberate. If cluster state traffic shared a link with production traffic, congestion or a misconfigured policy on that link could induce a false failover, or worse, a condition in which both members believe the peer has failed.

5. Hardware Components

Four physical devices constitute the infrastructure. Serial numbers below are as reported by the devices themselves.

Role	Model	Serial number	Function
FG-1	FortiGate 100E	FG100ETK20007401	Primary cluster member; FortiLink switch controller
FG-2	FortiGate 100E	FG100ETK20005486	Secondary cluster member; synchronised standby
sw1	FortiSwitch 1048E	FS1E48T422005405	MC-LAG peer; managed via FortiLink
sw2	FortiSwitch 1048E	FS1E48T422005438	MC-LAG peer; managed via FortiLink

The FortiGate 100E provides dedicated HA1 and HA2 interfaces on the front panel, which removes the need to sacrifice general-purpose ports for heartbeat duty. The FortiSwitch 1048E is a 48-port SFP+ chassis; its high port density and fibre interfaces made it suitable for both the FortiLink uplinks and the high-bandwidth Inter-Chassis Link required by MC-LAG.

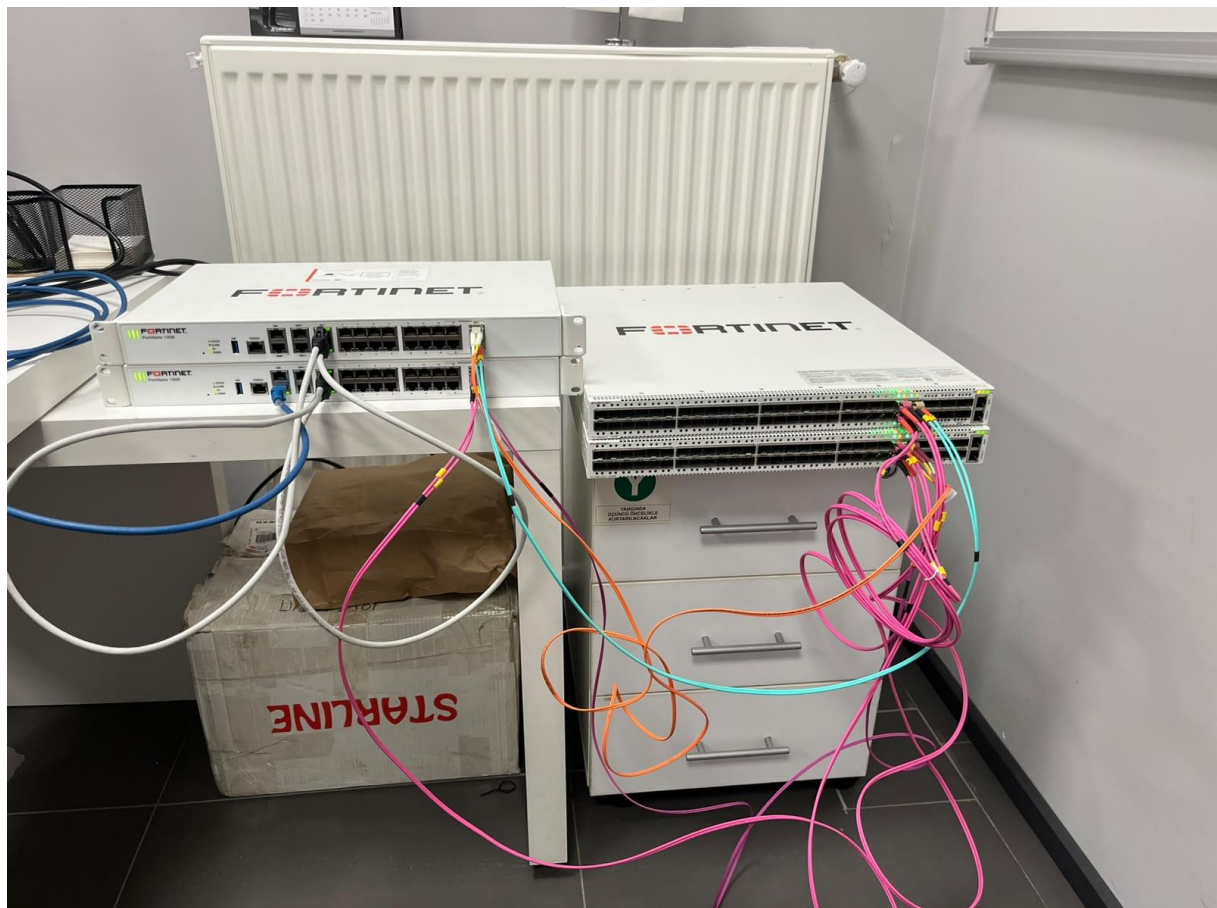


Figure 3. Laboratory installation. The FortiGate pair is mounted on the bench at left, the FortiSwitch pair at right, interconnected by copper heartbeat and fibre uplinks.

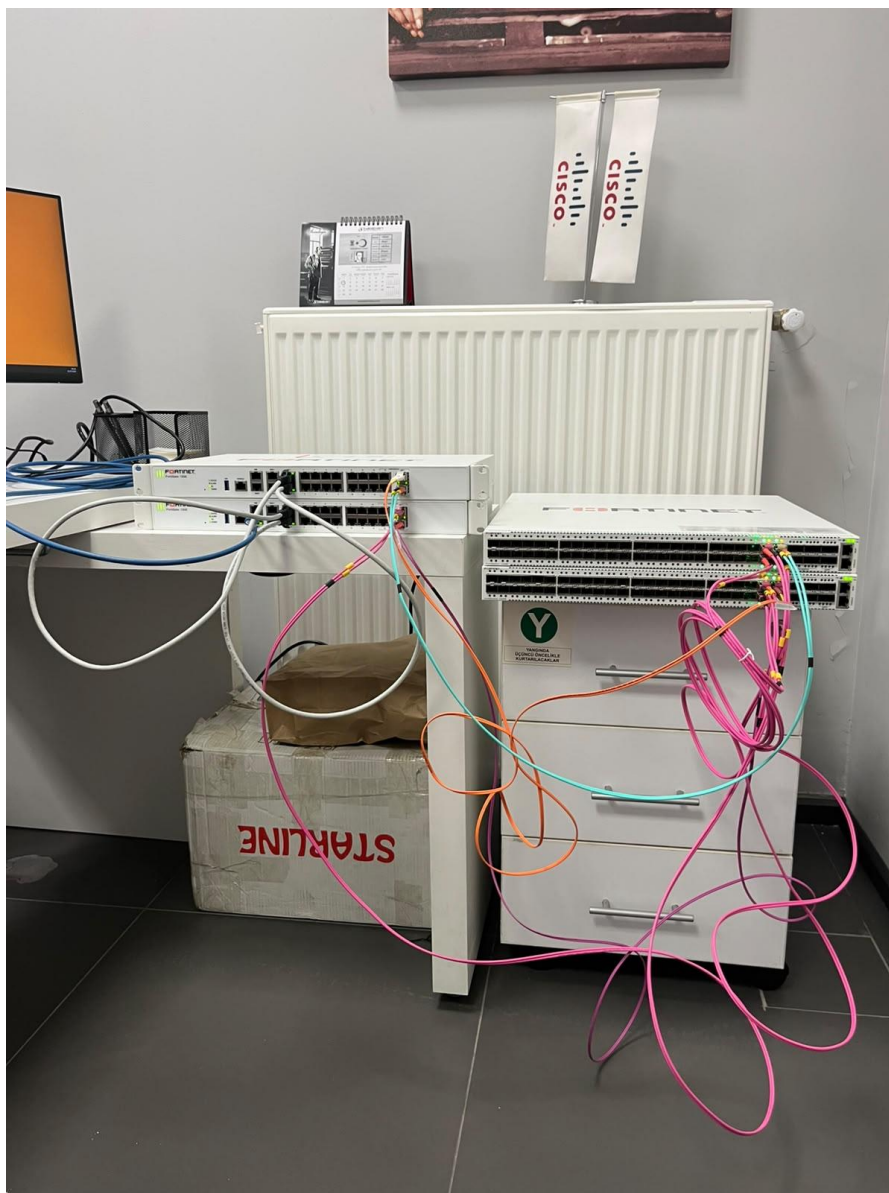


Figure 4. Alternative view of the installed hardware showing the complete cabling of the four-device topology.

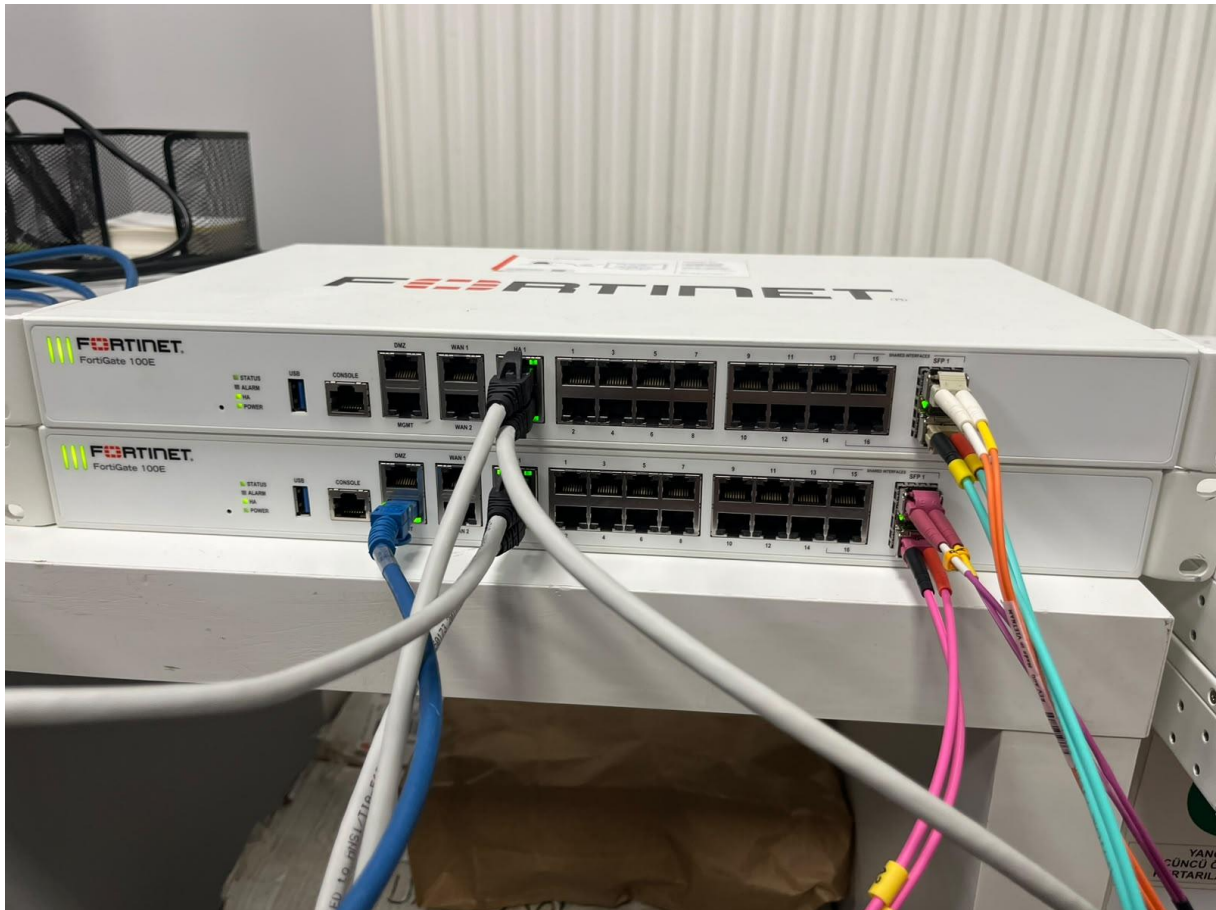


Figure 5. The two FortiGate 100E units. The HA1 interfaces are cabled between the chassis; the SFP uplinks at right carry the FortiLink aggregate to the switch pair.

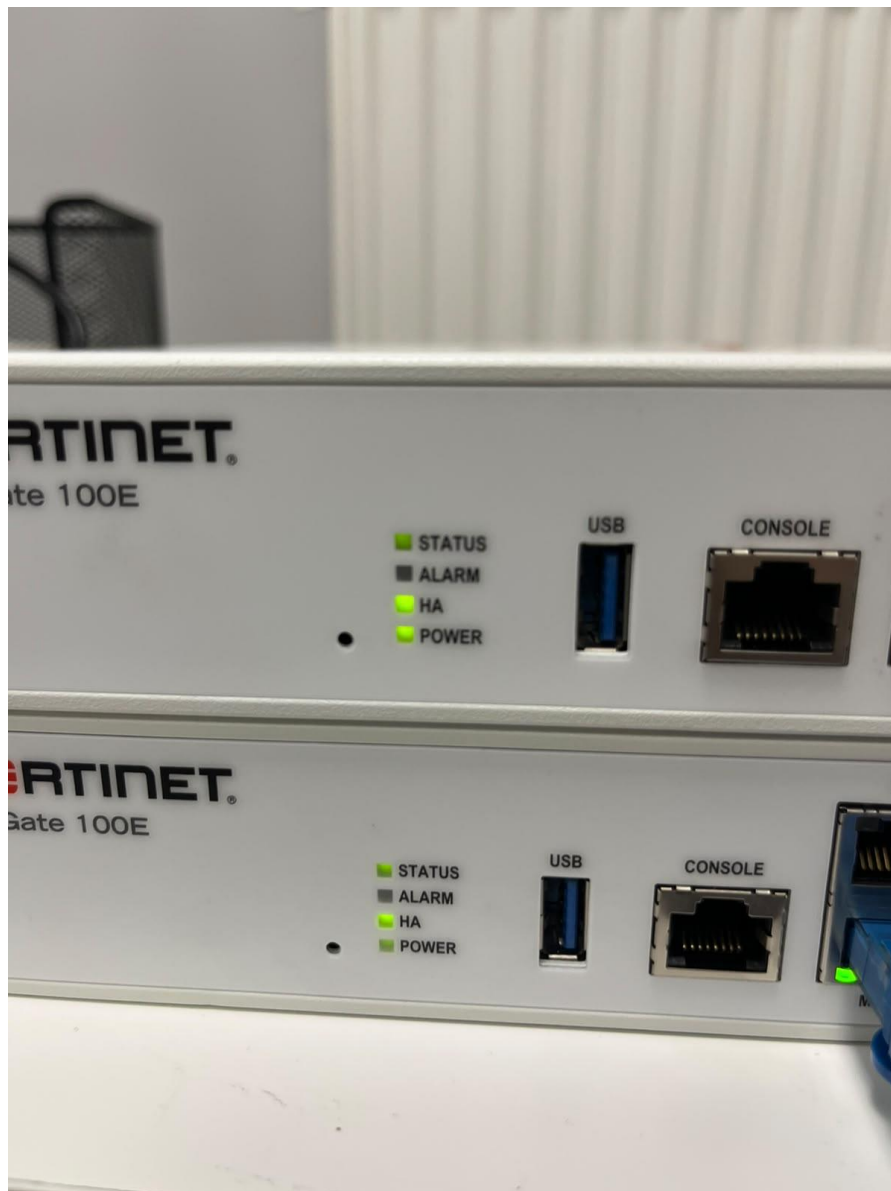


Figure 6. Front-panel status indication on both FortiGate units. The illuminated HA indicator on each chassis confirms cluster participation at the hardware level.

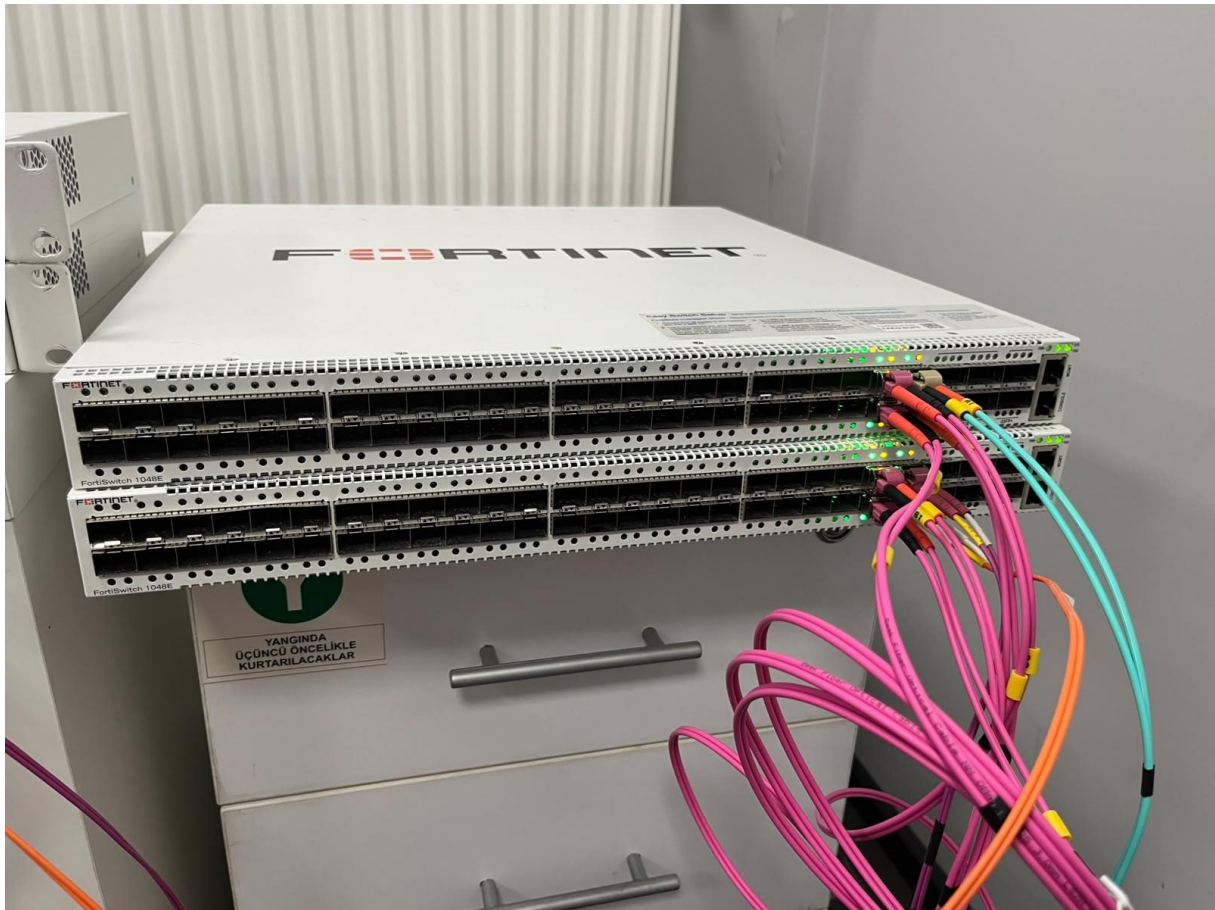


Figure 7. The two FortiSwitch 1048E chassis. The fibre bundle at right carries both the FortiLink uplinks to the firewall cluster and the Inter-Chassis Link between the two switches.

6. Technologies Used

6.1 FortiGate High Availability (Active-Passive)

FortiGate HA in Active-Passive mode operates two physical firewalls as one logical device. One member is elected primary and processes all traffic; the other maintains a synchronised copy of the configuration and session table and assumes the primary role if the active member fails. The cluster in this implementation runs with group ID 1, group name “admin”, and both members at the default priority of 128.

6.2 FortiLink

FortiLink is the management protocol by which a FortiGate assumes control of one or more FortiSwitch units. Under FortiLink the switches have no independently administered configuration; ports, VLANs and policies are defined on the FortiGate and pushed to the switches, which are presented in the firewall interface as managed extensions of the firewall itself.

6.3 MC-LAG (Multi-Chassis Link Aggregation)

MC-LAG allows two physically separate switch chassis to present themselves to a connected device as a single link aggregation partner. A device may therefore aggregate links across two switches and treat them as one logical interface, obtaining chassis-level redundancy without the connected device being aware that two chassis are involved.

6.4 Inter-Chassis Link (ICL)

The ICL is the link between the two MC-LAG peers over which they synchronise state and exchange keepalives. It is the mechanism that permits the two chassis to behave as one logical unit, and its integrity is a precondition for correct MC-LAG operation.

6.5 Firmware recovery via TFTP

FortiSwitch units expose a bootloader-level recovery path through which a firmware image can be loaded over TFTP via the console port. This procedure was required during the implementation and is described in Section 11.4.

7. Network Design

This section sets out the reasoning behind the design, and specifically why High Availability, FortiLink and MC-LAG were implemented together rather than selected individually. The central argument is that each of the three addresses a failure mode that the other two leave open.

7.1 Why HA alone is insufficient

An Active-Passive FortiGate cluster removes the firewall as a single point of failure. It does nothing for the layer beneath it. If both cluster members connect to the network through a single access switch, that switch inherits the role of the single point of failure that the cluster was deployed to eliminate. The availability of the system is then bounded by the availability of one switch, and the second firewall contributes nothing to the failure mode most likely to occur.

7.2 Why MC-LAG alone is insufficient

The symmetric argument applies to the switching layer. An MC-LAG pair removes the access switch as a single point of failure and, because both member links of an aggregate are forwarding simultaneously, does so without the idle standby capacity that a spanning-tree-based design imposes. But a redundant switch pair beneath a single firewall protects only against switch failure. Loss of the firewall remains total, and no amount of Layer 2 resilience compensates for it.

7.3 Why FortiLink is the element that unifies them

Deploying HA and MC-LAG together resolves both failure modes in isolation but introduces a third problem: administrative divergence. If the switches are managed independently of the firewalls, then switch configuration lives outside the cluster synchronisation domain. A failover transfers the firewall role to the secondary member, but the secondary member has no authority over the

switching layer and no synchronised copy of its configuration. The infrastructure survives the failure while losing coherent management of half of itself.

FortiLink resolves this by placing the switches inside the cluster's configuration domain. Switch configuration is authored on the FortiGate, and because that configuration is part of the cluster configuration, it is synchronised to the secondary member along with everything else. When the secondary is promoted it assumes control of the switch pair with a configuration already identical to that of the failed primary. FortiLink is therefore not a convenience feature in this design; it is what makes the firewall redundancy and the switch redundancy parts of one system rather than two.

7.4 The role of the ICL

MC-LAG requires that the two chassis agree on the state of the aggregates they jointly present. The ICL carries that state. In this implementation ports 45–46 on each chassis are dedicated to the ICL, with a keepalive interval of one second and a timeout of sixty seconds. Port 48 is configured as an egress-blocked port, which prevents traffic received across the ICL from being forwarded back out of an MC-LAG member port and thereby prevents the duplication of frames.

A further consequence of the ICL is the elimination of spanning-tree blocking on the redundant uplinks. Because the two chassis present one logical LAG partner, the parallel paths from the firewall cluster do not appear to Spanning Tree Protocol as a loop, and both uplinks remain in a forwarding state.

7.5 Coupling failover to FortiLink health

Two cluster settings were chosen to make the firewall tier responsive to conditions in the switching tier. Session pickup was enabled, so that the session table is synchronised to the standby member and established connections survive a role change rather than being reset. Interface monitoring was configured on the FortiLink aggregate, so that the cluster treats loss of FortiLink connectivity as a failure condition warranting failover.

The second of these is the more consequential design decision. Without interface monitoring, the cluster fails over only when a member ceases to function. A primary member that remains fully operational but has lost its path to the switching layer would retain the primary role while being unable to forward traffic, and the healthy secondary would remain passive. Monitoring the FortiLink aggregate closes this gap by making connectivity to the switch pair, rather than mere survival of the chassis, the criterion for holding the primary role.

Cluster override was left disabled. With override disabled the cluster does not preempt a functioning primary merely because a higher-priority member becomes available again, which avoids a second service interruption when a recovered member rejoins.

8. Implementation Process

The implementation was sequenced so that each stage established the preconditions required by the next. The ordering below was chosen deliberately and is not merely chronological.

8.1 Stage 1 — Physical installation

Both FortiGate units and both FortiSwitch units were installed and powered. Heartbeat cabling was run between the dedicated HA1 and HA2 interfaces of the two firewalls. Fibre uplinks were run from each firewall to both switches, and the Inter-Chassis Link was cabled between the two switch chassis.

8.2 Stage 2 — HA cluster formation

The Active-Passive cluster was configured and brought into synchronisation before any switch configuration was attempted. This ordering matters: because FortiLink and FortiSwitch configuration becomes part of the cluster configuration, establishing the cluster first ensures that every subsequent configuration change is synchronised to both members as it is made. Configuring the switching layer first would have required a later reconciliation of divergent configurations.

8.3 Stage 3 — FortiLink interface

The FortiLink interface was created as an 802.3ad aggregate over port15 and port16 and assigned the address 10.255.1.1/24 in dedicated FortiSwitch mode. Building FortiLink on an aggregate rather than a single physical port means the management and control channel to the switching layer is itself redundant.

8.4 Stage 4 — Switch discovery and authorisation

With FortiLink active, both FortiSwitch units were discovered and authorised into the management domain of the cluster, receiving addresses 10.255.1.2 and 10.255.1.3 on the FortiLink network.

8.5 Stage 5 — MC-LAG and ICL

The Inter-Chassis Link was established on ports 45–46 and the two switches were formed into an MC-LAG peer group. Only after the ICL was confirmed operational were the MC-LAG aggregates brought into service, since MC-LAG behaviour is undefined while the peers are unable to synchronise state.

9. Configuration Overview

This section presents the configuration as it stands on the implemented system. Only settings actually present on the devices are described.

9.1 High Availability

The cluster configuration was reviewed on the primary member:

```
show system ha
```

The cluster operates in a-p mode under group ID 1 and group name “admin”. Two dedicated heartbeat devices are configured, ha1 and ha2, at differing priorities so that one is preferred while the other remains available as a redundant heartbeat path. Session pickup is enabled, override is

disabled, and the FortiLink interface is monitored. The encrypted HA password has been masked in the figure below.

```
1FortiGate-100E # show system ha
config system ha
  set group-id 1
  set group-name "admin"
  set mode a-p
  set password ENC [REDACTED - HA password hash]

  set hbdev "ha1" 65 "ha2" 38
  set session-pickup enable
  set override disable
  set monitor "fortilink"
end
1FortiGate-100E #
```

Figure 8. HA cluster configuration on the primary member. Dual heartbeat devices, session pickup, and FortiLink interface monitoring are visible. The password hash has been redacted.

9.2 FortiLink interface

The FortiLink interface is an 802.3ad aggregate whose members are port15 and port16. Its addressing mode is set to “Dedicated to FortiSwitch” with the address 10.255.1.1/255.255.255.0, and automatic authorisation of connected devices is enabled. The split-interface option is disabled, which is the required setting for this topology: with split interface disabled the two member ports operate as a single aggregate toward the MC-LAG peer group rather than as two separate FortiLink paths.

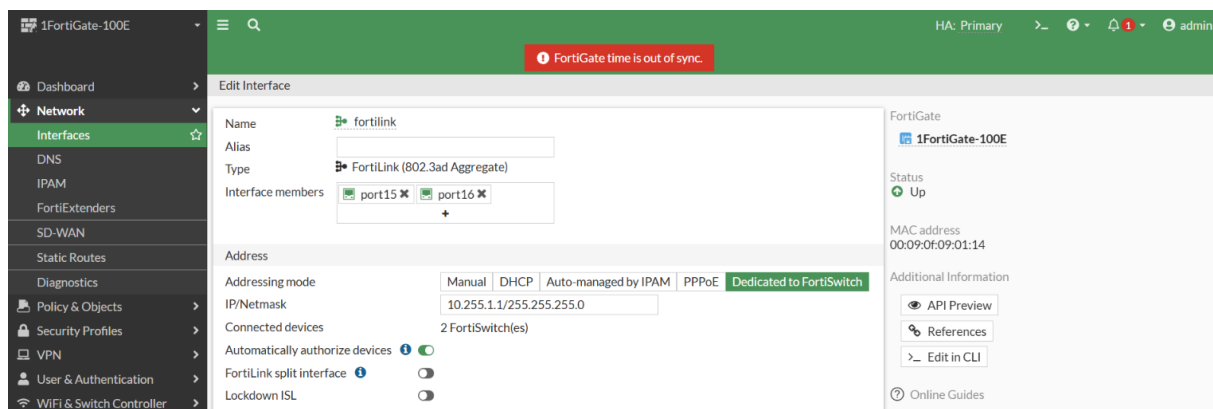
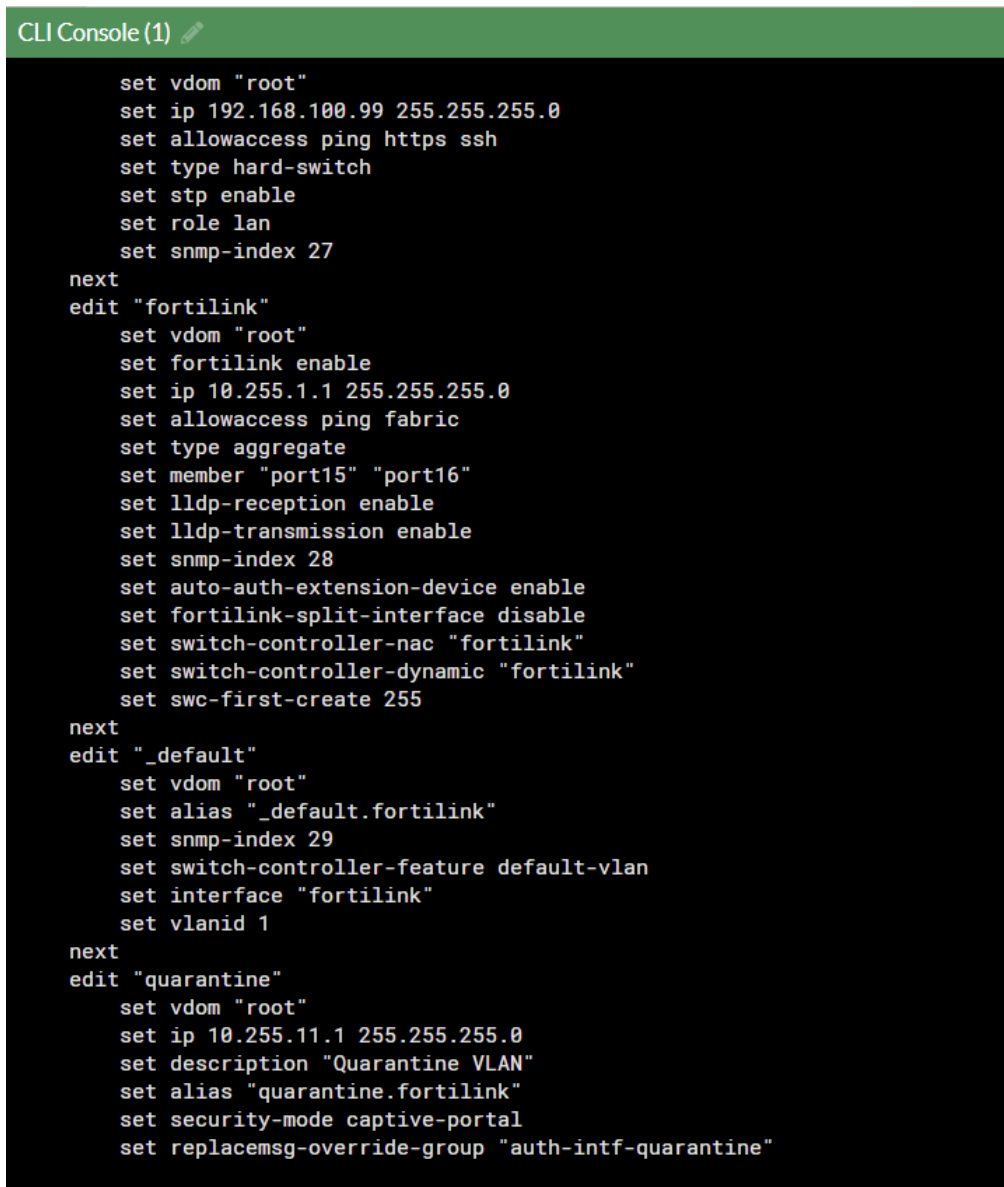


Figure 9. FortiLink interface configuration. The aggregate spans port15 and port16, carries 10.255.1.1/24 in dedicated FortiSwitch mode, and reports two connected FortiSwitch devices.

The same interface as represented in the configuration file:

```
show system interface
```



```
CLI Console (1)
set vdom "root"
set ip 192.168.100.99 255.255.255.0
set allowaccess ping https ssh
set type hard-switch
set stp enable
set role lan
set snmp-index 27
next
edit "fortilink"
set vdom "root"
set fortilink enable
set ip 10.255.1.1 255.255.255.0
set allowaccess ping fabric
set type aggregate
set member "port15" "port16"
set lldp-reception enable
set lldp-transmission enable
set snmp-index 28
set auto-auth-extension-device enable
set fortilink-split-interface disable
set switch-controller-nac "fortilink"
set switch-controller-dynamic "fortilink"
set swc-first-create 255
next
edit "_default"
set vdom "root"
set alias "_default.fortilink"
set snmp-index 29
set switch-controller-feature default-vlan
set interface "fortilink"
set vlanid 1
next
edit "quarantine"
set vdom "root"
set ip 10.255.11.1 255.255.255.0
set description "Quarantine VLAN"
set alias "quarantine.fortilink"
set security-mode captive-portal
set replacemsg-override-group "auth-intf-quarantine"
```

Figure 10. Interface configuration showing the FortiLink aggregate with LLDP enabled in both directions, automatic device authorisation, split interface disabled, and the associated default and quarantine VLANs.

LLDP transmission and reception are enabled on the aggregate; this is the mechanism by which FortiSwitch units are discovered on the FortiLink segment. A quarantine VLAN is provisioned at 10.255.11.1/255.255.255.0 with captive-portal security mode, providing an isolation segment for devices that fail authorisation.

10. Verification Process

Verification was performed through the FortiGate graphical interface and command line interface. Each item below states what was checked and what the observed state establishes. All figures in this section show device-reported state at the time of capture.

10.1 System status

```
get system status
```

The primary member reports FortiOS v7.2.13 build 1762, serial number FG100ETK20007401, operating in NAT mode with current HA mode reported as a-p, primary. This establishes the platform baseline and confirms that the device holds the primary role.

```
CLI Console (1) 
1FortiGate-100E # get system status
Version: FortiGate-100E v7.2.13,build1762,260128 (GA.M)
Security Level: High
Firmware Signature: certified
Virus-DB: 1.00000(2018-04-09 18:07)
Extended DB: 1.00000(2018-04-09 18:07)
AV AI/ML Model: 0.00000(2001-01-01 00:00)
IPS-DB: 6.00741(2015-12-01 02:30)
IPS-ETDB: 0.00000(2001-01-01 00:00)
APP-DB: 6.00741(2015-12-01 02:30)
FMWP-DB: 0.00000(2001-01-01 00:00)
INDUSTRIAL-DB: 6.00741(2015-12-01 02:30)
IPS Malicious URL Database: 1.00001(2015-01-01 01:01)
IoT-Detect: 0.00000(2022-08-17 17:31)
Serial-Number: FG100ETK20007401
BIOS version: 05000008
System Part-Number: P18827-04
Log hard disk: Not available
Hostname: 1FortiGate-100E
Private Encryption: Disable
Operation Mode: NAT
Current virtual domain: root
Max number of virtual domains: 10
Virtual domains status: 1 in NAT mode, 0 in TP mode
Virtual domain configuration: disable
FIPS-CC mode: disable
Current HA mode: a-p, primary
Cluster uptime: 51 minutes, 40 seconds
Cluster state change time: 2026-07-21 09:20:54
Branch point: 1762
Release Version Information: GA
System time: Tue Jul 21 10:03:37 2026
Last reboot reason: shutdown

1FortiGate-100E #
```

Figure 11. System status on the primary cluster member, confirming firmware version, serial number and current HA role.

10.2 HA cluster status

The cluster view in the graphical interface lists both members with their roles, priorities and synchronisation state. Both members report Synchronized at priority 128, with FG100ETK20007401 holding the primary role and FG100ETK20005486 the secondary.

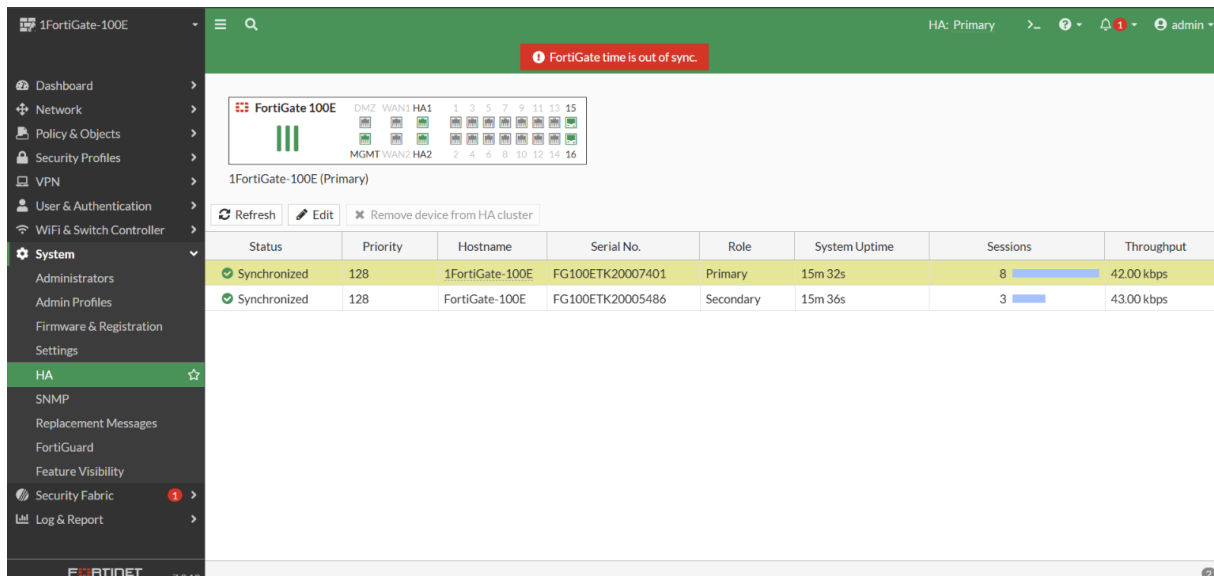


Figure 12. HA cluster view. Both members report a synchronised state, with primary and secondary roles correctly assigned.

get system ha status

The command line view reports HA Health Status OK and Mode HA A-P. It records the reason for the current primary election, states the configuration status of both members as in-sync, and shows both heartbeat devices ha1 and ha2 up at 1000 Mbps on both members. The FortiLink aggregate appears under monitored devices in an up state on both members, confirming that interface monitoring is active.

```
CLI Console (1)
1FortiGate-100E # get system ha status
HA Health Status: OK
Model: FortiGate-100E
Mode: HA A-P
Group Name: admin
Group ID: 1
Debug: 0
Cluster Uptime: 0 days 0:52:49
Cluster state change time: 2026-07-21 09:20:54
Primary selected using:
<2026/07/21 09:20:54> vcluster-1: FG100ETK20007401 is selected as the primary because the value 0 of link-failure + pingsvr-failure is less than peer member FG100ETK20005486.
<2026/07/21 09:12:20> vcluster-1: FG100ETK20007401 is selected as the primary because it's the only member in the cluster.
ses_pickup: enable, ses_pickup_delay=disable
override: disable
Configuration Status:
FG100ETK20007401(updated 0 seconds ago): in-sync
FG100ETK20007401 checksum dump: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15
FG100ETK20005486(updated 3 seconds ago): in-sync
FG100ETK20005486 checksum dump: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15
System Usage stats:
FG100ETK20007401(updated 0 seconds ago):
sessions=0, average-cpu-user/nice/system/idle=1%/0%/0%/98%, memory=25%
FG100ETK20005486(updated 3 seconds ago):
sessions=1, average-cpu-user/nice/system/idle=2%/0%/0%/97%, memory=24%
HBDEV stats:
FG100ETK20007401(updated 0 seconds ago):
ha1: physical/1000auto, up, rx-bytes/packets/dropped/errors=7510941/20159/0/0, tx=7677911/17614/0/0
ha2: physical/1000auto, up, rx-bytes/packets/dropped/errors=6258457/13093/0/0, tx=6231791/13092/0/0
FG100ETK20005486(updated 3 seconds ago):
ha1: physical/1000auto, up, rx-bytes/packets/dropped/errors=7667803/17591/0/0, tx=7500715/20138/0/0
ha2: physical/1000auto, up, rx-bytes/packets/dropped/errors=6220368/13068/0/0, tx=6249853/13075/0/0
MONDEV stats:
FG100ETK20007401(updated 0 seconds ago):
fortilink: aggregate/00, up, rx-bytes/packets/dropped/errors=18936486/26650/0/0, tx=2512290/14447/0/0
FG100ETK20005486(updated 3 seconds ago):
fortilink: aggregate/00, up, rx-bytes/packets/dropped/errors=4496750/7947/0/0, tx=66778/1745/0/0
Primary : 1FortiGate-100E , FG100ETK20007401, HA cluster index = 0
Secondary : FortiGate-100E , FG100ETK20005486, HA cluster index = 1
```

Figure 13. HA cluster status. Health is OK, both members are in-sync, both heartbeat devices are up on both members, and the monitored FortiLink aggregate is up.

diagnose sys ha status

The diagnostic view confirms the cluster operating parameters and the priority relationship between members: both carry a user priority of 128, and the serial number priority resolves the election deterministically in favour of FG100ETK20007401.

```

1FortiGate-100E # diagnose sys ha status
HA information
Statistics
    traffic.local = s:0 p:51835 b:30205756
    traffic.total = s:0 p:51832 b:30213781
    activity.ha_id_changes = 2
    activity.fdb = c:0 q:0

Model=100, Mode=2 Group=1 Debug=0
nvcluster=1, ses_pickup=1, delay=0

[Debug_Zone HA information]
HA group member information: is_manage_primary=1.
FG100ETK20007401:    Primary, serialno_prio=0, usr_priority=128, hostname=1FortiGate-100E
FG100ETK20005486:    Secondary, serialno_prio=1, usr_priority=128, hostname=FortiGate-100E

[Kernel HA information]
vcluster 1, state=work, primary_ip=169.254.0.1, primary_id=0
FG100ETK20007401:    Primary, ha_prio/o_ha_prio=0/0
FG100ETK20005486:    Secondary, ha_prio/o_ha_prio=1/1

1FortiGate-100E #

```

Figure 14. HA diagnostic status showing group membership, role assignment and the priority values governing primary election.

10.3 HA synchronisation

```
diagnose sys ha checksum cluster
```

Synchronisation was verified by comparing configuration checksums between the two members. Both report identical global, root and all-VDOM checksums, with the all checksum reading b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15 on each. Matching checksums are the authoritative confirmation that the two members hold the same configuration; the in-sync status reported elsewhere is a summary of this comparison.

```

1FortiGate-100E # diagnose sys ha checksum cluster

===== FG100ETK20007401 =====

is_manage_primary()=1, is_root_primary()=1
debugzone
global: 89 eb 90 f2 30 e8 f8 0a 3a 37 35 f2 45 52 55 41
root: 30 bd f4 c9 11 2c b5 65 47 f8 b3 fb ef c7 69 20
all: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15

checksum
global: 89 eb 90 f2 30 e8 f8 0a 3a 37 35 f2 45 52 55 41
root: 30 bd f4 c9 11 2c b5 65 47 f8 b3 fb ef c7 69 20
all: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15

===== FG100ETK20005486 =====

is_manage_primary()=0, is_root_primary()=0
debugzone
global: 89 eb 90 f2 30 e8 f8 0a 3a 37 35 f2 45 52 55 41
root: 30 bd f4 c9 11 2c b5 65 47 f8 b3 fb ef c7 69 20
all: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15

checksum
global: 89 eb 90 f2 30 e8 f8 0a 3a 37 35 f2 45 52 55 41
root: 30 bd f4 c9 11 2c b5 65 47 f8 b3 fb ef c7 69 20
all: b0 e2 3f 0a 19 9d cb 50 53 c4 0b 62 d4 96 17 15

1FortiGate-100E #

```

Figure 15. Configuration checksum comparison across cluster members. Identical values on both members confirm complete synchronisation.

10.4 FortiLink connectivity

execute switch-controller get-conn-status

Both switches are reported Authorized/Up on the fortilink interface, at addresses 10.255.1.2 and 10.255.1.3, with a managed switch count of 2 up and 0 down against a platform maximum of 32.

```

1FortiGate-100E # execute switch-controller get-conn-status
Managed-devices in current vdom root:

FortiLink interface : fortilink

```

SWITCH-ID	VERSION	STATUS	FLAG	ADDRESS	JOIN-TIME	NAME
FS1E48T422005405	v7.2.2 (419)	Authorized/Up	-	10.255.1.2	Tue Jul 21 09:15:06 2026	sw1
FS1E48T422005438	v7.2.9 (517)	Authorized/Up	-	10.255.1.3	Tue Jul 21 09:22:17 2026	sw2

```

Flags: C=config sync, U=upgrading, S=staged, D=delayed reboot pending, E=config sync error, 3=L3
Managed-Switches: 2 (UP: 2 DOWN: 0 MAX: 32)

1FortiGate-100E #

```

Figure 16. FortiSwitch connection status. Both units are authorised and up on the FortiLink interface with their assigned addresses and join times.

```
get switch-controller managed-switch
```

```
1FortiGate-100E # get switch-controller managed-switch
== [ FS1E48T422005405 ]
switch-id: FS1E48T422005405
== [ FS1E48T422005438 ]
switch-id: FS1E48T422005438

1FortiGate-100E #
```

Figure 17. Managed switch inventory, listing both FortiSwitch units by switch identifier.

10.5 Managed FortiSwitch status

The graphical inventory reports both switches online and identifies each by model and firmware version. Both units are FortiSwitch 1048E. The firmware versions differ between the two units, a condition addressed in Section 11.4.

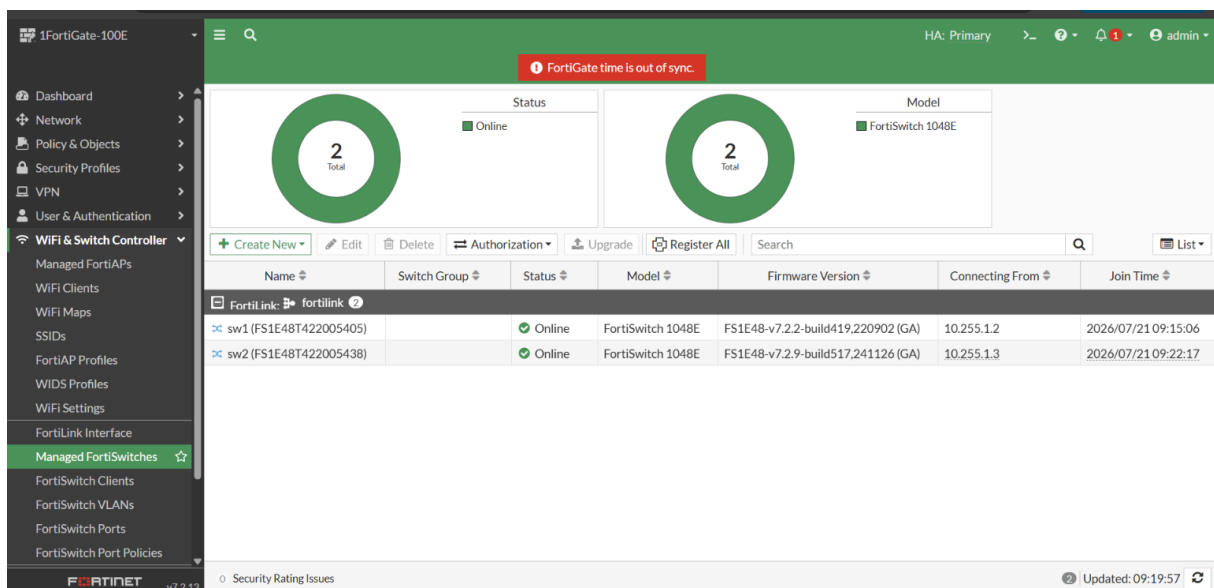


Figure 18. Managed FortiSwitch inventory. Both units report online status under the FortiLink interface, with model and firmware version for each.

The topology view renders the FortiLink stack and confirms the MC-LAG peer group relationship between the two switches, showing the uplink ports on each chassis and the ICL between them.

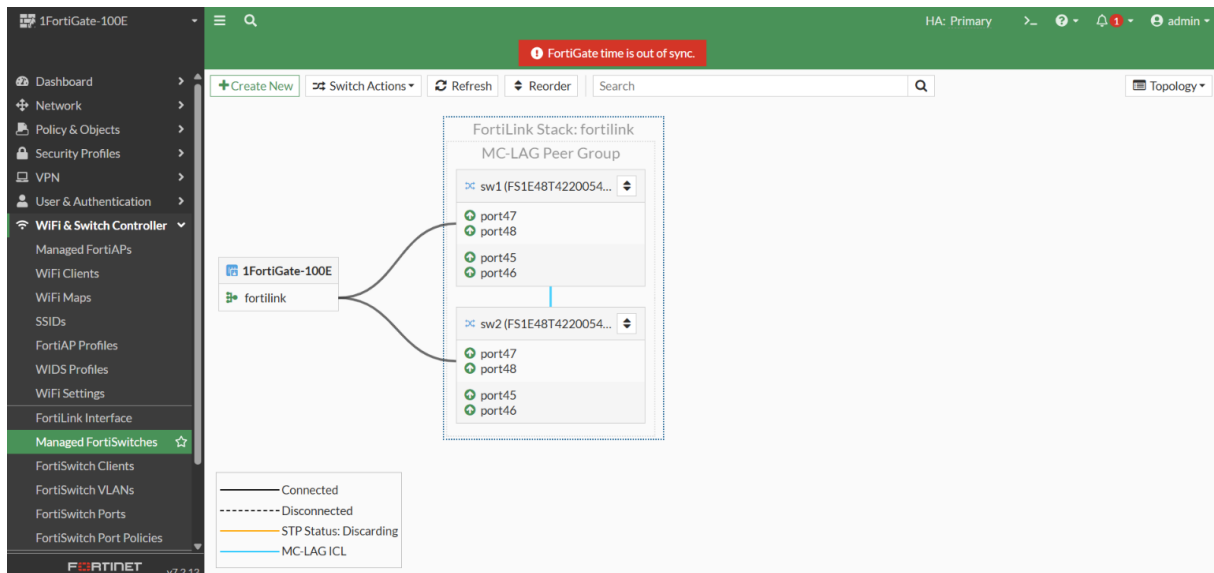


Figure 19. FortiLink topology view showing the MC-LAG peer group, the uplink ports on each chassis and the MC-LAG ICL between the two switches.

10.6 MC-LAG status

`diagnose switch mclag icl`

The ICL diagnostic confirms the peer relationship between the two chassis. Both peers are identified as FortiSwitch-1048E, the ICL is established on ports 45–46 with port 48 egress-blocked, and local and peer serial numbers correspond to the two installed switches. The keepalive counters show sustained bidirectional exchange, and split-brain is reported as disabled. The output additionally raises a peer software mismatch warning, discussed in Section 11.4.

```
sw2 # diagnose switch mclag icl
_FlInK1_ICL0_
  model          FortiSwitch-1048E
  peer model      FortiSwitch-1048E
  software        v7.2.9,build0517,241126 (GA)
  peer software    v7.2.2,build0419,220902 (GA) **peer software mismatch
  icl-ports       45-46
  egress-block-ports 48
  interface-mac    84:39:8f:12:08:6d
  local-serial-number FS1E48T422005438
  peer-mac         84:39:8f:11:fe:e3
  peer-serial-number FS1E48T422005405
  Local uptime     0 days 0h:24m:0s
  Peer uptime      0 days 0h:23m:0s
  MCLAG-STP-mac    84:39:8f:12:08:6c
  keepalive interval 1
  keepalive timeout 60
  dormant candidate Peer
  split-brain      Disabled

Counters
  received keepalive packets 701
  transmitted keepalive packets 697

sw2 #
```

Figure 20. MC-LAG Inter-Chassis Link status. The peer relationship is established on ports 45–46 with active keepalive exchange. Note the peer software mismatch warning.


```
diagnose switch mclag list
```

The MC-LAG list confirms that both cluster members are recognised as MC-LAG peers and that the aggregates are operational, with local LAG ports filtered for ICL traffic to prevent frame duplication across the inter-chassis path.

```
CLI Console (1) 

sw2 # diagnose switch mclag list
(*) - Using local system-id in LACP BPDU

G100ETK20007401(*)
-----
Local system ID          84:39:8f:12:08:9d
Peer system ID           84:39:8f:12:08:9d
Current system ID        84:39:8f:12:08:9d
Local ports              48
Peer ports               47-48
Local uptime             0 days  0h: 9m:19s
Peer uptime              0 days  0h:14m:56s
Local LAG is configured as LACP active.
Atleast one local LAG port is UP.
Peer LAG is configured as LACP active
Atleast one peer LAG port is UP,
local LAG ports are filtered for ICL traffic.
Updates sent to peer      564
Updates received from peer 563

G100ETK20005486(*)
-----
Local system ID          84:39:8f:12:08:9c
Current system ID        84:39:8f:12:08:9c
Local ports              47
Local uptime             0 days  0h: 0m: 0s
Local LAG is configured as LACP active.
Updates sent to peer      532
Updates received from peer 0
```

Figure 21. MC-LAG peer list showing system identifiers, member ports and update exchange between the peers.

10.7 FortiLink aggregate status

```
diagnose netlink aggregate name fortilink
```

The aggregate is reported up with two member ports. LACP operates in active mode at slow transmission rate, with an L4 distribution algorithm and a minimum link count of one. Both port15 and port16 report LACP state established and MUX state COLLECTING_DISTRIBUTING, which confirms that both members are actively carrying traffic rather than one standing by. Both members report the same partner system identifier, which is the expected result for an aggregate terminating on an MC-LAG peer group: the two physical chassis present themselves to the FortiGate as a single LACP partner.

```
CLI Console (1) 
1FortiGate-100E # diagnose netlink aggregate name fortilink
LACP flags: (A|P)(S|F)(A|I)(I|O)(E|D)(E|D)
(A|P) - LACP mode is Active or Passive
(S|F) - LACP speed is Slow or Fast
(A|I) - Aggregatable or Individual
(I|O) - Port In sync or Out of sync
(E|D) - Frame collection is Enabled or Disabled
(E|D) - Frame distribution is Enabled or Disabled

status: up
npu: y
flush: n
asic helper: y
oid: 84
ports: 2
link-up-delay: 50ms
min-links: 1
ha: master
distribution algorithm: L4
LACP mode: active
LACP speed: slow
LACP HA: enable
aggregator ID: 1
actor key: 17
actor MAC address: e0:23:ff:21:c4:7a
partner key: 17
partner MAC address: 84:39:8f:12:08:9d

member: port15
  index: 0
  link status: up
  link failure count: 0
  permanent MAC addr: e0:23:ff:21:c4:7a
  LACP state: established
  actor state: ASAIEE
  actor port number/key/priority: 1 17 255
  partner state: ASAIEE
  partner port number/key/priority: 1 17 255
  partner system: 2 84:39:8f:12:08:9d
```

Figure 22. FortiLink aggregate status. Both member ports report an established LACP state and are actively collecting and distributing traffic against a single logical partner.

11. Troubleshooting

Several faults were encountered during deployment. Each is documented below with the symptom observed, the diagnostic approach taken and the resolution applied.

11.1 FortiLink discovery

Switches connected to the FortiLink segment did not initially appear in the managed device inventory. The diagnostic approach was to work upward through the dependency chain rather than to alter configuration speculatively, since FortiLink discovery depends on several conditions that fail with the same visible symptom.

The conditions were verified in order: that the interface was of FortiLink type with addressing mode set to dedicated FortiSwitch operation rather than a conventional static address; that LLDP transmission and reception were enabled on the aggregate, since discovery depends on LLDP exchange; that the physical uplinks terminated on ports participating in the aggregate; and that the split-interface setting matched the intended topology. The authoritative check at each step was:

```
execute switch-controller get-conn-status
```

This command distinguishes the possible states directly. A switch absent from the listing has not been discovered, which indicates a physical or LLDP-layer problem. A switch present but not authorised has been discovered successfully, which relocates the problem to the authorisation stage and rules out the entire physical and discovery path. Using this distinction avoided investigating cabling for what was in fact an authorisation condition.

11.2 Switch authorisation

A discovered switch does not enter the managed domain until it is authorised. The FortiLink interface in this implementation has automatic authorisation of connected devices enabled, so discovered units are admitted without manual intervention; where automatic authorisation is not in effect, each unit must be authorised explicitly through the managed switch inventory.

The join times recorded for the two units differ: sw1 joined at 09:15:06 and sw2 at 09:22:17 on 21 July 2026. The interval between them corresponds to the additional work required on the second unit described in Section 11.4. The join sequence is visible in the connection status output shown in Figure 16.

11.3 Connectivity problems during deployment

Cluster stability issues during the build phase were diagnosed from the HA event history, which records state changes with timestamps and is the most direct evidence available for intermittent conditions:

```
diagnose sys ha history read
```

The history records heartbeat loss from member FG100ETK20005486 on both ha1 and ha2 at 13:38:20 and again at 16:29:42 on 17 July 2026. In both instances the cluster reverted to a single-member state and re-elected FG100ETK20007401 as primary. Loss on both heartbeat devices simultaneously indicates a condition affecting the peer member or both heartbeat paths together rather than the failure of an individual link, since independent link faults would not be expected to coincide.

The history also records link transitions on the FortiLink member ports during the build: port16 transitioned down and back up between 09:19:48 and 09:19:53 on 21 July, with the cluster re-forming as a two-member cluster at 09:20:54 when the peer rejoined. This sequence corresponds to the physical work being carried out on the FortiLink uplinks at that time and confirms that the cluster recovered without manual intervention.

```
CLI Console (1)
1FortiGate-100E # diagnose sys ha history read
version=1.1
HA state change time: 2026-07-21 09:20:54
message_count=50/512
<2026-07-21 09:21:08> hbdev ha2 link status changed: 0->1
<2026-07-21 09:21:08> port ha2 link status changed: 0->1
<2026-07-21 09:20:54> FG100ETK20007401 is elected as the cluster primary of 2 member
<2026-07-21 09:20:54> new member 'FG100ETK20005486' joins the cluster
<2026-07-21 09:20:53> hbdev ha1 link status changed: 0->1
<2026-07-21 09:20:53> port ha1 link status changed: 0->1
<2026-07-21 09:19:53> port port16 link status changed: 0->1
<2026-07-21 09:19:48> port port16 link status changed: 1->0
<2026-07-21 09:17:56> port port16 link status changed: 0->1
<2026-07-21 09:14:20> port fortilink link status changed: 0->1
<2026-07-21 09:14:18> port port15 link status changed: 0->1
<2026-07-21 09:13:21> port mgmt link status changed: 0->1
<2026-07-21 09:12:20> FG100ETK20007401 is elected as the cluster primary of 1 member
<2026-07-21 09:12:00> hatalk started
<2026-07-17 16:29:55> hatalk exited
<2026-07-17 16:29:42> hbdev ha2 link status changed: 1->0
<2026-07-17 16:29:42> hbdev ha1 link status changed: 1->0
<2026-07-17 16:29:42> port ha2 link status changed: 1->0
<2026-07-17 16:29:42> port ha1 link status changed: 1->0
<2026-07-17 16:29:42> FG100ETK20007401 is elected as the cluster primary of 1 member
<2026-07-17 16:29:42> heartbeats from FG100ETK20005486 are lost on all hbdev
<2026-07-17 16:29:42> member FG100ETK20005486 lost heartbeat on hbdev ha2
<2026-07-17 16:29:42> member FG100ETK20005486 lost heartbeat on hbdev ha1
<2026-07-17 15:11:31> port fortilink link status changed: 1->0
<2026-07-17 15:11:30> port port15 link status changed: 1->0
<2026-07-17 15:11:18> port port16 link status changed: 1->0
<2026-07-17 13:47:07> port mgmt link status changed: 0->1
<2026-07-17 13:46:49> hbdev ha2 link status changed: 0->1
<2026-07-17 13:46:49> port ha2 link status changed: 0->1
<2026-07-17 13:46:41> FG100ETK20007401 is elected as the cluster primary of 2 member
<2026-07-17 13:46:41> hbdev ha1 link status changed: 0->1
<2026-07-17 13:46:41> port ha1 link status changed: 0->1
<2026-07-17 13:46:41> new member 'FG100ETK20005486' joins the cluster
<2026-07-17 13:38:21> hbdev ha1 link status changed: 1->0
<2026-07-17 13:38:21> port ha1 link status changed: 1->0
```

Figure 23. HA event history. Heartbeat loss and cluster re-election events on 17 July are visible, together with the FortiLink port transitions and cluster re-formation on 21 July.

The general principle applied was to treat the event history as the primary source for intermittent faults. A status command reports the state of the system at the moment it is run and cannot distinguish a system that has been stable from one that has recently recovered; the history preserves that distinction.

11.4 Firmware recovery using TFTP

The ICL diagnostic reports a peer software mismatch between the two switch chassis. The installed versions differ: sw1 runs v7.2.2 build 419 and sw2 runs v7.2.9 build 517. Divergent firmware across MC-LAG peers is not a cosmetic condition. The peers must agree on the protocol behaviour governing state synchronisation across the ICL, and version differences introduce the risk of inconsistent handling of that state.

During the deployment one FortiSwitch unit became unreachable through its normal management path and could not be recovered through the standard upgrade mechanism, which requires a

functioning system to accept the image. Recovery was therefore performed through the bootloader, which is the appropriate tool when the operating system itself cannot be relied upon to complete an upgrade.

The procedure followed was:

- A console connection was established to the affected switch, since the bootloader is reachable only through the serial console and not over the network.
- The unit was power-cycled and interrupted during boot to enter the bootloader menu.
- A host on the same segment as the switch management interface was configured as a TFTP server and the firmware image placed in its root directory.
- The TFTP transfer option was selected from the bootloader menu and the server address, local address and image filename supplied.
- The image was transferred, written to flash, and the unit rebooted onto the recovered firmware.

Following recovery the unit rejoined the FortiLink domain and was authorised, which accounts for its later join time relative to the first switch. The engineering observation to record is that the mismatch warning remains present in the diagnostic output at the time of capture: recovery restored the unit to service but did not by itself align the two chassis on a common firmware version. Aligning them is identified as outstanding work in Section 14.1.

11.5 General configuration troubleshooting

Where configuration behaviour was uncertain, the approach taken was to establish device-reported state before making changes, rather than to reconfigure and observe the effect. Three checks were used as the standard sequence:

- **Checksum comparison** confirms that both cluster members hold identical configuration and isolates whether a fault is cluster-wide or specific to one member.
- **Connection status** distinguishes a switch that has not been discovered from one that has been discovered but not authorised, which are different faults with the same presentation.
- **Event history** separates a persistent fault from one that has already recovered, which a point-in-time status command cannot do.

Applying these before changing configuration avoided the failure mode in which speculative changes accumulate on a system whose actual state has not been established, leaving a configuration that is difficult to reason about even after the original fault is resolved.

12. Results

The implementation achieved the following verified states:

Objective	Verified state	Evidence
HA cluster operational	HA Health Status OK, mode HA A-P, roles correctly assigned	Figures 12, 13
Cluster synchronised	Both members in-sync with identical configuration checksums	Figure 15

Heartbeat redundancy	ha1 and ha2 both up on both members	Figure 13
FortiLink established	Aggregate up over port15 and port16 on 10.255.1.1/24	Figures 9, 22
Switches under central management	2 managed switches, 2 up and 0 down	Figures 16, 18
Switch authorisation	Both units Authorized/Up at 10.255.1.2 and 10.255.1.3	Figure 16
MC-LAG peer group formed	ICL established on ports 45–46 with active keepalive exchange	Figures 19, 20
Both uplinks forwarding	Both aggregate members in COLLECTING_DISTRIBUTING state	Figure 22

The infrastructure meets the objectives set out in Section 3. No single device in the topology is a cut vertex of the forwarding graph: the failure of either firewall leaves a synchronised peer able to assume the primary role with the session table intact, and the failure of either switch leaves a peer chassis carrying the aggregate. The switching layer is administered as part of the cluster configuration and is therefore synchronised to both members.

One condition remains outstanding. The two switch chassis are running different firmware versions and the ICL diagnostic continues to report a peer software mismatch. The MC-LAG relationship is established and operational in this state, but the condition should not be regarded as acceptable for a production deployment.

13. Conclusion

This project implemented a redundant network infrastructure on physical Fortinet hardware, combining an Active-Passive FortiGate 100E cluster with a FortiLink-managed FortiSwitch 1048E MC-LAG pair. The implementation was verified through the graphical and command line interfaces of the firewall cluster, and the resulting state confirms cluster synchronisation, central switch management and an established inter-chassis relationship between the two switches.

The principal engineering conclusion concerns the relationship between the three technologies rather than any one of them individually. High Availability, MC-LAG and FortiLink each address a distinct failure mode, and a design that deploys any one alone remains bounded by the failure mode it does not cover. High Availability without redundant switching is bounded by the switch; MC-LAG without firewall redundancy is bounded by the firewall; and both together, without FortiLink, produce an infrastructure that survives component failure while losing coherent management of the switching layer at precisely the moment a failover occurs. The decision to bind the switching layer into the cluster configuration domain, and to make the cluster monitor its own path to that layer, is what makes the design a single resilient system.

A final observation concerns the limits of what this report establishes. The redundancy implemented here has been verified as a configured and reported state: the devices report a synchronised cluster, an established inter-chassis relationship and both uplinks forwarding. That is not the same as

demonstrated behaviour under fault. The distinction is material, and the work required to close it, together with the other development identified during the implementation, is set out in Section 14.

14. Future Improvements

The following developments are identified as the appropriate continuation of this work. They are ordered by engineering priority rather than by effort: the first two address gaps in what the current implementation has established, while the remainder extend its operational capability.

14.1 Alignment of FortiSwitch firmware versions

The two switch chassis are running different firmware versions and the ICL diagnostic reports a peer software mismatch. This is the only defect in the implemented infrastructure that is visible in the verification output, and it should therefore be addressed first. Both chassis should be brought onto a common version, with the upgrade performed one chassis at a time so that the surviving peer continues to carry the aggregate throughout. The ICL diagnostic provides the confirmation that the condition has cleared.

14.2 Comprehensive failover testing

The redundancy described in this report has been verified as configuration and reported state, not as observed behaviour under fault. A structured test programme would close this gap by exercising each failure mode the design claims to cover: loss of the primary cluster member, loss of an individual heartbeat interface, loss of a single FortiLink member port, loss of an entire switch chassis, and loss of the ICL. Each scenario should record whether the role transition occurred as intended and whether established sessions survived it, which is the specific claim that session pickup is intended to support. Until such testing is carried out, the resilience of the design remains an argument from configuration rather than a demonstrated property.

14.3 FortiAnalyzer integration

The cluster currently retains logs locally, and the primary member reports no log hard disk available. Integrating a FortiAnalyzer would provide centralised, off-box log retention for both cluster members and both managed switches. Beyond storage capacity, the operational benefit is that log data would survive the failure of the device that generated it, which is a precondition for investigating any incident in which a device was itself the fault.

14.4 Long-term monitoring and log analysis

The HA event history proved to be the most informative diagnostic source during this implementation, but it is a bounded buffer and is lost on reboot. Continuous monitoring through SNMP, together with retained logging, would allow cluster state changes, heartbeat interruptions and switch join events to be correlated over time rather than inspected as isolated snapshots. Intermittent faults are characterised by their distribution over time, and a point-in-time diagnostic is structurally unable to reveal that distribution.

14.5 Performance measurement under traffic load

No performance characterisation was carried out during this implementation, and no throughput, latency or failover timing figures are claimed anywhere in this report. Establishing that behaviour would require generating representative traffic across the MC-LAG aggregate and measuring forwarding performance under load, the distribution of traffic across aggregate member links, and the duration of the traffic interruption during a controlled failover. These measurements would convert the qualitative availability argument made in Section 7 into quantified figures suitable for capacity planning and for service level definition.

15. References

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